Ollscoil Mhá Nuad
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SEMESTER 2
2025-2026

CS417
Numerical Computation

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Time allowed: 2 hours

Answer all questions.

Not all questions carry equal marks.

Please round to **six decimal places** unless specifically told to do otherwise.

Instructions

	Yes	No	N/A
Formulae and Tables book allowed *(i.e., available on request)*	X		
Formulae and Tables book required *(i.e., distributed prior to exam commencing)*		X	
Statistics Tables and Formulae allowed *(i.e., available on request)*	X		
Statistics Tables and Formulae required *(i.e., distributed prior to exam commencing)*		X	
Dictionary allowed *(supplied by the student)*		X	
Non-programmable calculator allowed	X		
Students required to write in and return the exam question paper		X	

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[40 marks]

1 (a) Given the two orthogonal matrices U and V below [10 marks]

$$U = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \; V = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

Construct the block matrix,

$$\mathrm{M} = \frac{1}{\sqrt{2}} \begin{pmatrix} U & U \\ V & -V \end{pmatrix}$$

i) Compute the inverse of M, Minv.
ii) Compute the Frobenius norm, F_U, of U.
iii) Compute the Frobenius norm, F_{Minv}, of Minv.

(b) Consider the following integral. [10 marks]

$$Un = \int_1^e x(\ln(x))^n \, dx$$

And its related recurrence relation

$$Un = \frac{1}{2}(e)^2 - \frac{n}{2} U_{n-1}$$

i) Comment on the numerical stability of this relation.
ii) Calculate U_0 exactly.
iii) Given that $U_{20}<0.333$, estimate U_{18} in a numerically stable way. Justify your answer.

(c) The values of an unknown function, f(x), are tabulated below for given x values, [10 marks]

x	f(x)
1.75	0.168172
1.77	0.102296
1.83	-0.107335
1.85	-0.179240

Use an inverse interpolation method to estimate the local root of f(x). Note a graphical estimate will not receive any marks.

Please see Next Page for part (d) →

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1 (d) Consider the predator, y, prey, x, system, [10 marks]

$$\frac{dx}{dt} = x - a\,x^2 - \frac{b\,x\,y}{x+6}$$

$$\frac{dy}{dt} = -c\,y + \frac{dx\,y}{x+6}$$

Where a=0.2, b=2, c=0.25 and d=1.0.

Given the initial population (x0, y0) = (9, 5) and using a step size of h=1 year, estimate the populations, (x, y), after 1 year. Justify your answer with appropriate calculations.

Note: The following may be used in your answer,

$$f3 = \begin{bmatrix} -10.497599 \\ 1.632303 \end{bmatrix}$$

$$f4 = \begin{bmatrix} 2.465945 \\ -3.864128 \end{bmatrix}$$

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[30 marks]

2 (a) Consider the matrix. [20 marks]

$$A = \begin{pmatrix} 2 & 3 & 0 & 0 \\ 3 & 2 & 3 & 0 \\ 0 & 3 & 2 & 3 \\ 2 & 0 & 3 & 2 \end{pmatrix}$$

Let the matrix $M = N^{-1}P$ where N and P are the Jacobi splitting of A.

i) Calculate ||M|| where ||.|| is the ill conditioning norm of a matrix.

ii) What does this tell you about the iterative convergence of the original matrix A?

iii) Construct the characteristic equation of the M and show it is the polynomial.

$$p(\lambda) = \lambda^4 - 6.75\,\lambda^2 + 1.6875$$

iv) Hence or otherwise determine if the original matrix A is suitable for Jacobi Iteration. Justify your answer with appropriate calculations.

(b) The following (x, f(x)) data have been taken from an unknown function, f. [10 marks]

x	f(x)
-3.00	0.300025
-2.14	0.215043
-1.29	0.151800
-0.43	0.653292
0.43	3.346708
1.29	3.848200
2.14	3.784957

This data has been interpolated with a cubic spline in python whose coefficients are.

```
[[ 0.028501 -0.107989  1.265715 -2.370954  1.256714 -0.071987]
 [ 0.        0.073289 -0.204396  3.050299 -3.046441  0.185109]
 [-0.120085 -0.057265 -0.169642  2.269703  2.273009 -0.179561]
 [ 0.300025  0.215043  0.151800  0.653292  3.346708  3.848200]]
```

You are asked for the maximum value of f(x) over the interval [0.43. 1.29].

i) Write down the component of the Spline covering this interval.

ii) Use the spline to calculate this maximum value.

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[30 marks]

3 (a) The equations of motion for three unmanned arial vehicles (UAVs) are, [20 marks]

$$F_1 = z - 5 + 0.5x - 0.0125x^2 + 0.6y^2$$
$$F_2 = z - 2 - 0.5tan^{-1}\left(\frac{y}{x}\right)$$
$$F_3 = z - 1 - 0.8\,sin\left(\frac{y - 0.5\cos(x)}{2}\right)$$

Testers are interested in finding the coordinates **x** = (x, y, z) of a point of collision between the three UAVs.
Given initial values for **x0** = (x, y, z) = (-1, 2.5, 2), use an appropriate method to estimate **x1**, the next iteration of x. Explain your approach.

Note: The partial Jacobian at **x0** is,

$$\mathrm{J}(\mathbf{x0}) = \begin{pmatrix} 0.274999 & 3.000060 & 1 \\ ******* & 0.068963 & 1 \\ 0.074090 & ******* & 1 \end{pmatrix}$$

(b) Consider the matrix, [10 marks]

$$A = \begin{pmatrix} 7 & 3 & 0 & 0 \\ 3 & 7 & 0 & 0 \\ 0 & 0 & 11 & 1 \\ 0 & 0 & -2 & 8 \end{pmatrix}$$

i) Given an initial vector,
v0= (0.482850, 0.575555, 0.080612, 0.655053),
use an appropriate numerical method to compute the next iteration, **v1**, which is used to estimate the smallest eigenvalue of A.

ii) Given the algorithm terminates with
z = (-1.76776692e-01, 1.76776699e-01, 0, 0)
and did not change sign, estimate the smallest eigenvalue of A. Explain your reasoning.

iii) Briefly explain how you could speed up this algorithm using eigenvalue shifting. Use appropriate notation.

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CS417 Data Sheet

Polynomial interpolation error for the data set
$(x_1, y_1), (x_2, y_2), ...(x_N, y_N).$

$$f(x) - P_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(\xi_x) w_n(x)$$

where

$$w_n(x) = \prod_{j=1}^{n} (x - x_j)$$

$$\min(x_1, ..., x_n, x) < \xi_x < \max(x_1, ..., x_n, x)$$

The Lagrangian Interpolating polynomial for the data set $(x_1, y_1), (x_2, y_2), ...(x_N, y_N).$

$$P_N(x) = \sum_{k=1}^{N} y_k L_K(x),$$

$$L_k(x) = \prod_{\substack{j=1 \\ j \neq k}}^{N} \frac{x - x_j}{x_k - x_j}$$

Newton divided difference formulae

$$f_1(x_k) = \frac{f_0(x_{k+1}) - f_0(x_k)}{x_{k+1} - x_k}$$

$$f_2(x_k) = \frac{f_1(x_{k+1}) - f_1(x_k)}{x_{k+2} - x_k}$$

$$f_j(x_k) = \frac{f_{j-1}(x_{k+1}) - f_{j-1}(x_k)}{x_{k+j} - x_k}$$

Neville's Method

$$P_k[x_i, x_{i+1}, x_{i+2}, \ldots\ldots, x_{i+k-1}, x_{i+k}] =$$

$$\frac{(x - x_{i+k}) P_{k-1}[x_i, x_{i+1}, \ldots, x_{i+k-1}] + (x_i - x) P_{k-1}[x_{i+1}, \ldots, x_{i+k}]}{x_i - x_{i+k}}$$

Starting with $P_0[x_i] = f(x_i)$

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CRT

Given $x = a_1 \bmod m_1$
$x = a_2 \bmod m_2$
$x = a_3 \bmod m_3$
……
$x = a_n \bmod m_n$

Then $x = \sum_{i=1}^{n} a_i M_i N_i \bmod (M)$

where $M = m_1 m_2 \ldots\ldots m_n$

$M_i = \frac{M}{m_i}$

and $N_i = M_i^{-1} \bmod m_i$

Relaxation w opt

$$\frac{2}{1+\sqrt{1-\left(\frac{\Delta x^{(k+p)}}{\Delta x^{(k)}}\right)^{1/p}}}$$

Central Difference formulae

$\underline{O(h^2)}$

$$f'(x_0) \approx \frac{f_1 - f_{-1}}{2h}$$

$$f''(x_0) = \frac{f_1 - 2f_0 + f_{-1}}{h^2} + E_2(f,h)$$

$$|E_2(f,h)| \leq \frac{4\mu}{h^2} + \frac{Mh^2}{12} \quad \text{where} \quad |f^{(4)}(x)| < M$$

Richardson Extrapolation

$$G = \frac{2^p g(h_1/2) - g(h_1)}{2^p - 1}$$

Power method
With Euclidean Scaling

The normalised power sequence

$$x_0,\ x_1 = \frac{Ax_0}{\|Ax_0\|}, x_2 = \frac{Ax_1}{\|Ax_1\|}, \ldots\ldots, x_k = \frac{Ax_k}{\|Ax_k\|}$$

converges to a unit dominant eigenvector and the sequence

$$Ax_1.x_1, Ax_2.x_2, Ax_3.x_3, \ldots\ldots, Ax_k.x_k,$$

converges to the dominant eigenvalue λ.

Gold Search

$|b-a|R^n = \varepsilon$

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Taylor's theorem

Let f(x) have n+1 continuous derivatives on [a, b] and let x, $x_0 \in$ [a, b]. Then

$f(x) = p_n(x) + R_n(x)$

where $p_n(x) = \sum_{k=0}^{n} \frac{(x-x_0)^k}{k!} f^{(k)}(x_0)$

and

$$R_n(x) = \frac{(x-x_0)^{n+1}}{(n+1)!} f^{(n+1)}(\xi_x)$$

Where ξ_x is a point between x and x_0.

Ridder's method

Given f, x_1 and x_2,
Output new interval [x_3, x_4]

Where $x_3 = (x_1 + x_2)/2$

$f_i = f(x_i)$

$s = \sqrt{f_3^2 - f_1 f_2}$

$x_4 = x_3 \pm (x_3 - x_1)\frac{f_3}{s}$

where $\begin{cases} + \; if \; f_1 > f_2 \\ - \; if \; f_1 < f_2 \end{cases}$

HouseHolder Matrix

$Q = I - \frac{\boldsymbol{u}\boldsymbol{u}^T}{H} \; where \; H = \frac{|\boldsymbol{u}|^2}{2}$

HouseHolder Reduction

(First iteration)

$$A \rightarrow \begin{bmatrix} A_{11} & (\boldsymbol{Q}\boldsymbol{x})^T \\ \boldsymbol{Q}\boldsymbol{x} & \boldsymbol{Q}\boldsymbol{A}'\boldsymbol{Q} \end{bmatrix}$$

Where $\mathbf{x} = [A_{12}, A_{13}, \ldots\ldots, A_{1n}]^T$

$k = \pm|\boldsymbol{x}|$ where $\begin{cases} + \; if \; x_1 > 0 \\ - \; if \; x_1 < 0 \end{cases}$

$u = [k + x_1, x_2, \ldots\ldots, x_n]^T$

$Qx = [-k, 0, 0, \ldots\ldots, 0]^T$

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Sturm Sequence of an n x n
Tridiagonal symmetric matrix
Main diagonal d lower diagonal c

$$P_0(\lambda) = 1$$
$$P_1(\lambda) = d_1 - \lambda$$
$$P_i(\lambda) = (d_i - \lambda)P_{i-1}(\lambda) - c_{i-1}^2 P_{i-2}(\lambda), \quad i = 2, 3, \ldots\ldots, n$$

Runge Kutta

$$\boldsymbol{y_{k+1}} = \boldsymbol{y_k} + \frac{h(\boldsymbol{f_1} + 2\boldsymbol{f_2} + 2\boldsymbol{f_3} + \boldsymbol{f_4})}{6}$$

where,

$$\boldsymbol{f}_1 = F(x_k, \boldsymbol{y_k})$$

$$\boldsymbol{f}_2 = F\left(x_k + \frac{h}{2}, \boldsymbol{y_k} + \frac{h}{2}\boldsymbol{f}_1\right)$$

$$\boldsymbol{f}_3 = F\left(x_k + \frac{h}{2}, \boldsymbol{y_k} + \frac{h}{2}\boldsymbol{f}_2\right)$$

$$\boldsymbol{f}_4 = F(x_k + h, \boldsymbol{y_k} + h\boldsymbol{f}_3)$$